MAT 221 - CALCULUS I
Lesson 8-9: - Trigonometric substitution

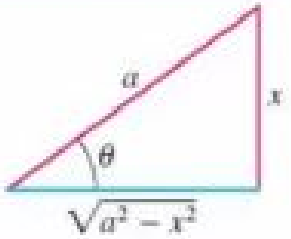
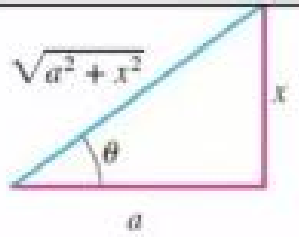
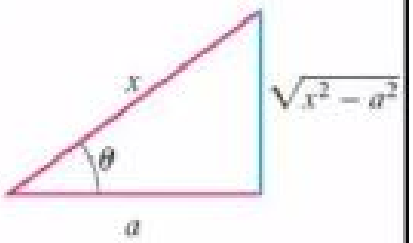
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Integrals Involving $\sqrt{a^2-x^2}$, $\sqrt{a^2+x^2}$, $\sqrt{x^2-a^2}$

The objective with trigonometric substitution is to eliminate the radical in the integrand. See the table below.

Expression	substitution	Identity	domain	triangle
$\sqrt{a^2-x^2}$	$x=a\sin\theta$	$1-\sin^2\theta=\cos^2\theta$	$-\frac{\pi}{2}\le\theta\le\frac{\pi}{2}$	
$\sqrt{a^2+x^2}$	$x=a\tan\theta$	$1+\tan^2\theta=\sec^2\theta$	$-\frac{\pi}{2}<\theta<\frac{\pi}{2}$	
$\sqrt{x^2-a^2}$	$x=a\sec\theta$	$\sec^2\theta-1=\tan^2\theta$	$0\le\theta<\frac{\pi}{2}$ or $\pi\le\theta<\frac{3\pi}{2}$	

Example 1

Find

(a) $\int \sqrt{9-x^2}\,dx$ (b) $\int x^2\sqrt{1-x^2}\,dx$ (c) $\int \frac{1}{\sqrt{36-x^2}}\,dx$

(d) $\int \frac{1}{\sqrt{4+x^2}}\,dx$ (e) $\int \frac{1}{\sqrt{25x^2-4}}\,dx, \quad x > \frac{2}{5}$

Solution

(a) Let $x = 3\sin\theta$ and $dx = 3\cos\theta\,d\theta$ then

$$\int \sqrt{9-x^2}\,dx = \int \sqrt{9-9\sin^2\theta}\cdot 3\cos\theta\,d\theta$$

$$= 9\int \cos^2\theta\,d\theta = \frac{9}{2}\int (1+\cos 2\theta)\,d\theta$$

$$= \frac{9}{2}\left(\theta + \frac{1}{2}\sin 2\theta\right) + C$$

$$= \frac{9}{2}\left[\sin^{-1}\left(\frac{x}{3}\right) + \frac{x}{3}\sqrt{1-\frac{x^2}{9}}\right] + C$$

since $x = 3\sin\theta \quad \therefore \sin\theta = \frac{x}{3}, \quad \cos\theta = \sqrt{1-\frac{x^2}{9}}$ and

$\sin 2\theta = 2\sin\theta\cos\theta = \frac{2x}{3}\sqrt{1-\frac{x^2}{9}}$

(b)-(d) In class

(e) $\sqrt{25x^2-4} = \sqrt{25\left(x^2-\frac{4}{25}\right)} = 5\sqrt{x^2-\left(\frac{2}{5}\right)^2}$ Letting $x = \frac{2}{5}\sec\theta$ and $dx = \frac{2}{5}\sec\theta\tan\theta\,d\theta$ then

$$\sqrt{x^2-\left(\frac{2}{5}\right)^2} = \sqrt{\frac{4}{25}\sec^2\theta - \frac{4}{25}} = \sqrt{\frac{4}{25}(\sec^2\theta-1)}$$

$$= \sqrt{\frac{4}{25}\tan^2\theta} = \frac{2}{5}\tan\theta$$

$$\therefore \int \frac{dx}{\sqrt{25x^2-4}} = \int \frac{dx}{5\sqrt{x^2-\left(\frac{2}{5}\right)^2}} = \int \frac{\frac{2}{5}\sec\theta\tan\theta\,d\theta}{5\left(\frac{2}{5}\tan\theta\right)}$$

$$= \frac{1}{5}\int \sec\theta\,d\theta = \frac{1}{5}\ln\left|\sec\theta+\tan\theta\right| + C$$

$$= \frac{1}{5}\ln\left|\frac{5x}{2} + \frac{\sqrt{25x^2-4}}{2}\right| + C.$$

Since $\sec\theta = \frac{5x}{2}$ and $\tan\theta = \sqrt{\sec^2\theta-1} = \sqrt{\left(\frac{5x}{2}\right)^2-1} = \frac{\sqrt{25x^2-4}}{2}$.

Integrals of the form $\int \frac{1}{a + b\sin^2 x + c\ \cos^2 x} dx$

Given

$$\int \frac{1}{a + b\sin^2 x + c\ \cos^2 x} dx,$$

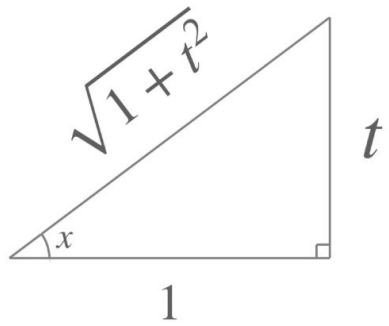
let $t = \tan x$ $dt = \sec^2 x\, dx$,

$$\frac{dx}{dt} = \frac{1}{\sec^2 x} = \frac{1}{1 + \tan^2 x}$$

therefore

$$\frac{dx}{dt} = \frac{1}{1 + t^2} \Rightarrow dx = \frac{1}{1 + t^2} dt$$

Using the right triangle



we have that $\sin x = \frac{t}{\sqrt{1+t^2}}$ and $\cos x = \frac{1}{\sqrt{1+t^2}}$.

Example 2

Find

(a) $\int \frac{1}{3+\cos^2 x}\,dx$ (b) $\int \frac{1}{2\cos^2 x+1}\,dx$

Solution

(a) Let $t = \tan x$, $dx = \frac{1}{1+t^2}dt$ and $\cos x = \frac{1}{\sqrt{1+t^2}}$

$3+\cos^2 x = 3+\frac{1}{1+t^2} = \frac{4+3t^2}{1+t^2}$ so that

$$\int \frac{1}{3+\cos^2 x}\,dx = \int \frac{1+t^2}{4+3t^2}\cdot\frac{1}{1+t^2}\,dt = \int \frac{1}{4+3t^2}\,dt$$

letting $A = 2$ and $z = \sqrt{3}t$, thus $dz = \sqrt{3}dt$ we have

$$\int \frac{1}{4+3t^2}\,dt = \frac{1}{\sqrt{3}}\int \frac{1}{2^2+z^2}\,dz = \frac{1}{\sqrt{3}}\cdot\frac{1}{2}\tan^{-1}\left(\frac{z}{2}\right)+C$$

$$= \frac{1}{2\sqrt{3}}\tan^{-1}\left(\frac{\sqrt{3}}{2}t\right)+C$$

$$= \frac{1}{2\sqrt{3}}\tan^{-1}\left(\frac{\sqrt{3}}{2}\tan x\right)+C$$

(b) Practice exercise

Integrals of the form $\int \frac{1}{a + b\sin x + c\ \cos x} dx$

Given

$$\int \frac{1}{a + b\sin x + c\ \cos x} dx,$$

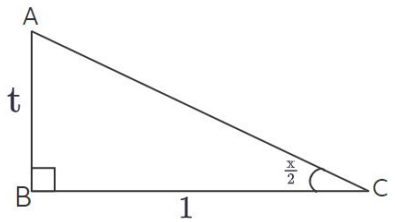
let $t = \tan \frac{x}{2}$ $dt = 2\sec^2 \frac{x}{2}\, dx$,

$$\frac{dx}{dt} = \frac{2}{\sec^2 \frac{x}{2}} = \frac{2}{1 + \tan^2 \frac{x}{2}}$$

therefore

$$\frac{dx}{dt} = \frac{2}{1 + t^2} \Rightarrow dx = \frac{2}{1 + t^2} dt$$

Using the right triangle



we have that $\sin \frac{x}{2} = \frac{t}{\sqrt{1+t^2}}$ and $\cos \frac{x}{2} = \frac{1}{\sqrt{1+t^2}}$. By double angle properties $\sin x = 2\sin \frac{x}{2} \cos \frac{x}{2} = \frac{2t}{1+t^2}$ and $\cos x = 2\cos^2 \frac{x}{2} - 1 = \frac{1-t^2}{1+t^2}$

Example 3

Find

(a) $\int \frac{1}{5+4\cos x}\,dx$ (b) $\int \frac{1}{3\sin x+4\cos x}\,dx$

Solution

(a) Let $t=\tan\frac{x}{2}$, $dx=\frac{2}{1+t^2}dt$ and $\cos x=\frac{1-t^2}{1+t^2}$

$5+4\cos x=5+\frac{4(1-t^2)}{1+t^2}=\frac{9+t^2}{1+t^2}$ so that

$$\int \frac{1}{5+4\cos x}\,dx=\int \frac{2}{9+t^2}\,dt$$

letting $A=3$ and $z=\sqrt{5}t$, thus $dz=\sqrt{5}dt$ we have

$$\int \frac{2}{9+t^2}\,dt=\int \frac{1}{3^2+t^2}\,dt=\frac{2}{3}\tan^{-1}\left(\frac{t}{3}\right)+C$$
$$=\frac{2}{3}\tan^{-1}\left(\frac{\tan(x/2)}{3}\right)+C$$

(b) Practice exercise